AnyAttack: Towards Large-scale Self-supervised Adversarial Attacks on Vision-language Models

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Abstract

Targeted adversarial attacks on vision-language models (VLMs) pose serious security threats, yet existing methods require either ground-truth labels or explicit target supervision, severely limiting their scalability and real-world applicability. In this paper, we introduce ***AnyAttack****, a self-supervised framework that removes this dependency by learning a universal adversarial noise generator without any labels or target text guidance. AnyAttack operates in two stages: (1)* pre-training *a decoder on the LAION-400M dataset using a contrastive loss to map CLIP image embeddings to adversarial perturbations, and (2)* fine-tuning *the decoder on downstream datasets with task-specific objectives. At inference, a single forward pass produces an adversarial image that steers any target VLM towards the embedding of a chosen target. Extensive experiments on MSCOCO, Flickr30K, and SNLI-VE demonstrate that AnyAttack substantially outperforms state-of-the-art baselines across image-text retrieval, multimodal classification, and image captioning tasks. AnyAttack also successfully attacks commercial VLMs including Google Gemini and OpenAI GPT-4o, achieving attack success rates of 31% and 38%, respectively, far exceeding all baselines. Beyond strong attack performance, AnyAttack is significantly more efficient in memory and runtime compared to iterative optimization-based methods, making it the first practical, large-scale foundation model for targeted adversarial noise generation.*

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1. Introduction

Vision-language models (VLMs) [12–14, 20] have achieved remarkable performance across a diverse spectrum of multimodal tasks, including image-text retrieval, visual question answering, and image captioning. As VLMs are increasingly deployed in safety-critical applications, understanding their vulnerability to adversarial attacks becomes essential.

Adversarial attacks on neural networks have been extensively studied in unimodal settings [9, 17, 24]. Translating these attacks to the multimodal domain is non-trivial, because VLMs must simultaneously process and align visual and textual modalities. *Targeted* adversarial attacks are particularly dangerous: they aim to manipulate a model into producing a specific, attacker-chosen output, rather than simply inducing any incorrect prediction. For VLMs, this means crafting an adversarial image that causes the model to associate it with an entirely different semantic concept.

Existing targeted attacks on VLMs [15, 32, 35, 37] share a critical limitation: they all require some form of *explicit supervision* to specify the attack target. As illustrated in Figure 2, prior methods either rely on ground-truth class labels or require a designated target image at attack time. Label-based methods are infeasible in open-vocabulary multimodal settings, while target-image-guided methods require iterative per-sample optimization, making them computationally expensive and not scalable to real-world deployment with millions of images.

We ask: *Can we learn to generate adversarial perturbations at scale, without any explicit target labels or guidance?*

We answer affirmatively with **AnyAttack**, a self-supervised adversarial attack framework. The core insight is to reframe targeted adversarial attack generation as a *foundation model pre-training problem*. Specifically, we train a perturbation generator F that maps any image embedding $z = E(x)$ to a perturbation $\delta = F(z)$, such that adding δ to a *random carrier image* x_r causes the resulting adversarial example $x_r + \delta$ to be perceived by a VLM as semantically similar to x. Crucially, this pre-training does not require any labels or text supervision: the self-supervised contrastive objective [4, 10] aligns the noised carrier embeddings to the original image embeddings.

This *pre-train then fine-tune* paradigm [3] for adversarial noise offers several compelling advantages. First, pre-training on the large-scale LAION-400M dataset [23] allows the generator to capture diverse, transferable visual features that generalize across models and tasks. Second, fine-tuning with lightweight task-specific objectives enables rapid adaptation to downstream applications. Third, inference requires only a single forward pass through the generator, in stark contrast to the hundreds or thousands of gradi-

ent steps required by iterative attacks, dramatically reducing both memory usage and runtime.

Our contributions are summarized as follows:

- We propose **AnyAttack**, the first self-supervised, label-free framework for large-scale targeted adversarial attacks on VLMs, which trains a universal perturbation generator without any label or text guidance.
- We introduce a pre-train then fine-tune paradigm for adversarial noise generation, using a contrastive pre-training objective on LAION-400M and task-specific fine-tuning on downstream datasets.
- We demonstrate that AnyAttack achieves strong cross-model and cross-task transferability, significantly outperforming all baselines on image-text retrieval, multimodal classification, and image captioning tasks, while also successfully attacking commercial VLMs (Gemini and GPT-4o).
- AnyAttack is substantially more efficient in both memory and runtime compared to iterative optimization baselines.

2. Related Work

Vision-Language Models. Large-scale VLMs have shown tremendous success by jointly learning visual and linguistic representations. CLIP [20] and ALIGN [12] train dual-encoder models via contrastive learning on massive noisy image-text pairs, yielding powerful zero-shot capabilities. BLIP [13] and BLIP-2 [14] further bridge the gap between image encoders and language models through bootstrapped pre-training. More recently, instruction-tuned models such as InstructBLIP [5] and MiniGPT-4 [38] enable flexible multimodal dialogue. These models' widespread deployment makes understanding their adversarial vulnerabilities critically important.

Adversarial Attacks. Adversarial attacks were first identified by Szegedy *et al.* [24] and further popularized by FGSM [9] and PGD [17]. Transfer-based black-box attacks [6, 7, 28, 30] generate adversarial examples on a surrogate model that transfer to target models without access to their parameters. Generative adversarial perturbation methods [18, 22] learn a feed-forward generator that amortizes the per-sample optimization, enabling faster inference. Recent work has explored improving transferability through input transformations [25, 29, 39], ensemble attacks, and self-supervised surrogate training [26].

Adversarial Attacks on VLMs. Adversarial attacks on VLMs present unique challenges due to the multimodal nature of these models. Co-Attack [34] jointly perturbs image and text modalities in a white-box setting. AdvCLIP [37] generates downstream-agnostic universal adversarial patches for CLIP-like models. Set-level Guidance Attack (SGA) [15], also referred to as SU in our experiments, proposes set-level objectives to boost transferability of targeted attacks. VLAttack [32] fuses image and text perturbations for cross-model black-box attacks. Attack-VLM [35] systematically evaluates transfer-based targeted attacks on large VLMs. The sample-agnostic SASD attack [36] explores universal perturbations via multimodal decision boundaries. More recently, diffusion-based methods [31] and ICLR approaches [16] have further pushed the boundaries of adversarial transferability.

Safety of Large Language and Vision-Language Models. The security of large models has become a critical research area. GCG [40] demonstrates universal adversarial suffixes for aligned language models. In the multimodal domain, visual adversarial prompts [19], bi-modal adversarial prompts [33], and indirect instruction injection [1] expose fundamental alignment vulnerabilities in deployed VLMs. Our work is complementary, focusing on the targeted attack setting where an adversary seeks to manipulate VLM outputs toward attacker-controlled semantics.

Self-Supervised Learning. Self-supervised representation learning methods such as MoCo [10], SimCLR [4], MAE [11], and BEiT [2] have demonstrated that rich visual representations can be learned from unlabeled data. AnyAttack draws inspiration from contrastive pre-training to learn adversarial perturbation generators without label supervision.

3. Proposed Method: AnyAttack

3.1. Problem Formulation

Let $E(\cdot)$ denote the image encoder of a surrogate VLM (CLIP ViT-B/32 in our case), and let $x_r \in \mathbb{R}^{3\times224\times224}$ denote a *carrier image* drawn from an independent distribution. Given a target image x with embedding $z = E(x) \in \mathbb{R}^{512}$, we seek a perturbation generator $F : \mathbb{R}^{512} \to \mathbb{R}^{3\times224\times224}$ such that:

$$x_{\text{adv}} = \text{clip}(x_r + F(z),\, 0, 1), \quad \|F(z)\|_\infty \le \varepsilon, \tag{1}$$

and $E(x_{\text{adv}}) \approx E(x)$ for various encoders E. Table 1 compares AnyAttack's formulation with prior work, highlighting that AnyAttack is the only approach that does not require explicit target supervision.

The key distinction is that existing methods minimize the loss between the adversarial example and a given target (label y_t or image x_t), which requires explicit supervision. AnyAttack instead aligns $\delta + x_r$ to the original source image x in a self-supervised manner.

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Strategy	Formulation
Target label supervision	$\min \mathcal{L}(f_s(x+\delta), y_t), \quad \text{s.t. } y_t \neq y$
Target image supervision	$\min \mathcal{L}(f_s(x+\delta), f_s(x_t)), \quad \text{s.t. } x_t \neq x$
AnyAttack (ours)	$\min \mathcal{L}(f_s(\delta + x_r), f_s(x))$

Table 1. The formulation of different attack strategies. The existing strategies rely on explicit target supervision, whereas our AnyAttack is unsupervised.

3.2. AnyAttack Architecture

The perturbation decoder F takes a 512-dimensional CLIP embedding as input and produces a spatial perturbation via the following architecture. An initial linear projection maps the embedding to a $256 \times 14 \times 14$ feature map. This is then passed through a series of alternating residual blocks and bilinear upsampling blocks, progressively upsampling from spatial resolution $14 \rightarrow 28 \rightarrow 56 \rightarrow 112 \rightarrow 224$. Each residual block consists of Conv(3×3) + BatchNorm + LeakyReLU + EfficientAttention. Each upsampling block consists of Nearest $\times 2$ + Conv(3×3) + BatchNorm + LeakyReLU. A final Conv($16 \rightarrow 3$) layer produces the perturbation, and the L_∞ constraint $\|\delta\|_\infty \leq \varepsilon = 16/255$ is enforced by clamping.

3.3. Pre-training on LAION-400M

Objective. We pre-train F on LAION-400M [23] using a self-supervised contrastive objective. For a mini-batch of images $\{x_i\}_{i=1}^B$, we apply K augmentations (with $K = 5$) by shuffling images within duplicated mini-batches, forming pairs (x_i, x_j) where x_i and x_j are different images. Carrier images x_r are drawn from the same batch. The pre-training loss is InfoNCE:

$$\mathcal{L}_{\text{Pre}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\text{sim}(E(x_{r,i} + F(E(x_i))),\, E(x_i))/\tau)}{\sum_{j=1}^{B} \exp(\text{sim}(E(x_{r,j} + F(E(x_i))),\, E(x_j))/\tau)} \tag{2}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity. The temperature τ is exponentially annealed from $\tau_0 = 1.0$ to $\tau_{\text{final}} = 0.07$ over $T = 10{,}000$ steps.

Training Details. The encoder E (CLIP ViT-B/32) is frozen throughout. We train for 520,000 steps using AdamW with learning rate 10^{-4}, weight decay 0.01, and CosineAnnealingWarmRestarts with $T_0 = 5{,}000$. Training uses batch size 600 per GPU on 3× A100 80GB GPUs with mixed precision. The K-augmentation strategy shuffles images within a duplicated mini-batch of size $K \times B$, creating diverse carrier-target pairs without needing labels.

3.4. Fine-tuning on Downstream Datasets

After pre-training, F is fine-tuned on downstream task datasets for 20 epochs with batch size 800. Random carrier images are drawn from the ImageNet-1K validation set. We employ two task-specific objectives:

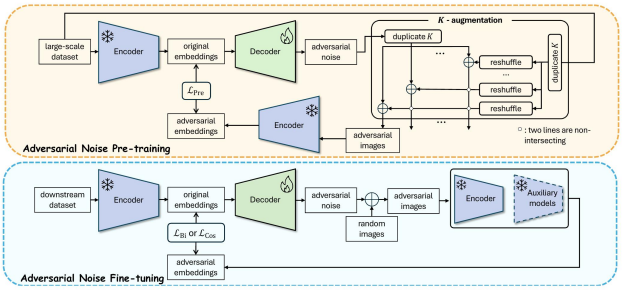

Figure 1. Overview of AnyAttack. The framework consists of a self-supervised pre-training stage on LAION-400M and a task-specific fine-tuning stage. At inference, a single forward pass produces the adversarial example.

Cosine Alignment Loss ($\mathcal{L}_{\text{Cos}}$): Used for general tasks such as classification and captioning:

$$\mathcal{L}_{\text{Cos}} = 1 - \text{sim}(E(x_r + F(E(x))),\, E(x)). \tag{3}$$

Bidirectional Contrastive Loss ($\mathcal{L}_{\text{Bi}}$): Used for retrieval tasks, with fixed $\tau = 0.07$:

$$\mathcal{L}_{\text{Bi}} = \mathcal{L}_{\text{i2t}} + \mathcal{L}_{\text{t2i}}, \tag{4}$$

where $\mathcal{L}_{\text{i2t}}$ and $\mathcal{L}_{\text{t2i}}$ are standard image-to-text and text-to-image InfoNCE losses.

Auxiliary Encoders. To improve cross-model transferability, we augment fine-tuning with two auxiliary frozen encoders beyond the CLIP ViT-B/32 surrogate: (1) a torchvision ViT-B/16 pretrained on ImageNet-1K at 224×224, and (2) EVA-02 Large Patch14 [8] at 448×448. The losses from all encoders are summed with equal weight. This ensemble supervision leads to substantially improved attack success rates on held-out VLMs (denoted as "w/ Aux" variants).

3.5. Inference

At inference time, given a target image x, the adversarial example is produced by a single forward pass:

$$\delta = F(E(x)), \quad x_{\text{adv}} = \text{clip}(x_r + \delta, 0, 1), \tag{5}$$

where x_r is a randomly chosen carrier image. This contrasts sharply with prior methods that require hundreds of iterative gradient steps per sample. An overview of the full AnyAttack framework is illustrated in Figure 1.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate on three benchmarks: MSCOCO (image-text retrieval and image captioning), Flickr30K

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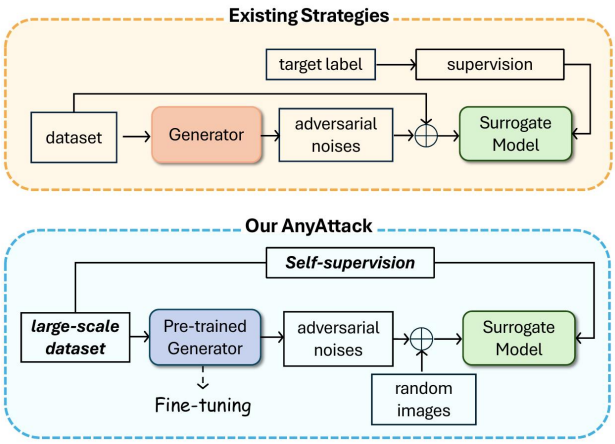

Figure 2. Comparison of existing targeted adversarial attack strategies and the proposed self-supervised AnyAttack. Prior methods require explicit target supervision (labels or target images), while AnyAttack operates without any supervision.

(image-text retrieval), and SNLI-VE (multimodal classification). For each dataset, we use the top 1,000 images as the target pool. Carrier images are drawn from the top 1,000 images of the ImageNet-1K [21] validation set.

Target VLMs. We evaluate transferability to: CLIP ViT-B/16, ViT-L/14, and ViT-L/14×336 [20]; BLIP [13]; BLIP-2 [14]; InstructBLIP [5]; and MiniGPT-4 [38]. We also evaluate on commercial VLMs: Google Gemini 1.5 Flash and OpenAI GPT-4o mini.

Baselines. We compare against: **AttackVLM-ii** and **AttackVLM-it** [35] (image-to-image and image-to-text transfer-based attacks); **SU-Cos** and **SU-MSE** (set-level guidance attacks [15] adapted with cosine and MSE losses); **SASD-WS-Cos** and **SASD-WS-MSE** [36] (sample-agnostic attacks adapted with cosine and MSE losses).

Evaluation Metrics. For image-text retrieval, we report recall at rank 1, 5, 10 (TR@k and IR@k) and their average (R@Mean). For multimodal classification (SNLI-VE), we report accuracy. For image captioning, we report SPICE, BLEU-1/4, METEOR, ROUGE-L, and CIDEr. For commercial VLMs, the attack success rate (ASR) is measured as the percentage of adversarial images rated as "highly/partially relevant" to the target text by the model.

Perturbation Budget. All attacks use L_∞ budget $\varepsilon = 16/255$.

4.2. Image-Text Retrieval on MSCOCO

Table 2 reports retrieval performance on MSCOCO. AnyAttack consistently and substantially outperforms all baselines across all three CLIP architectures. The best variant, *AnyAttack-Bi w/ Aux*, achieves R@Mean scores of 35.88, 33.66, and 33.18 for ViT-B/16, ViT-L/14, and ViT-L/14×336, respectively. This represents improvements of approximately 15–16 points over the best baseline (SASD-WS-MSE) for ViT-B/16 and even larger gains for ViT-L/14 and ViT-L/14×336.

The results reveal several important trends. First, the bidirectional contrastive loss ($\mathcal{L}_{\text{Bi}}$) consistently outperforms the cosine loss ($\mathcal{L}_{\text{Cos}}$) for retrieval, which is expected given the symmetric nature of image-text matching. Second, auxiliary encoder supervision provides substantial gains: *AnyAttack-Bi w/ Aux* outperforms *AnyAttack-Bi* by 9–16 R@Mean points, demonstrating that diverse surrogate ensemble supervision is key to cross-model transferability. Third, AttackVLM achieves near-zero performance, confirming that per-sample iterative attacks with limited steps fail to transfer. Notably, the performance gap between AnyAttack and all baselines widens for larger CLIP variants (ViT-L/14 and ViT-L/14×336), suggesting that strong generalization to higher-capacity encoders requires the pre-training diversity that iterative optimization with limited steps cannot provide.

4.3. Multimodal Classification on SNLI-VE

Table 3 reports results on the SNLI-VE dataset. *AnyAttack-Cos w/ Aux* achieves 44.8% accuracy, nearly doubling the best baseline (SASD-WS-MSE: 24.8%) and far exceeding AttackVLM (≤6.5%). Without auxiliary encoders, *AnyAttack-Cos* achieves 17.5%, underscoring the critical role of auxiliary supervision for classification transferability.

4.4. Image Captioning

Table 4 shows captioning results on MSCOCO for four generative VLMs. *AnyAttack-Cos w/ Aux* consistently achieves the best CIDEr scores: 20.3 for InstructBLIP (vs. 11.8 for SASD-WS-MSE), 13.3 for BLIP-2 (vs. 9.5), 13.0 for BLIP (vs. 12.4), and 5.3 for MiniGPT-4 (vs. 2.8). Gains are also consistent across SPICE, BLEU, METEOR, and ROUGE-L. These results demonstrate that AnyAttack generalizes to autoregressive generation models, causing them to produce captions that describe the adversarially targeted content rather than the carrier image.

4.5. Attacks on Commercial VLMs

We evaluate AnyAttack on commercial VLMs using a 100-image test set, prompting each model with "Evaluate the relationship between the given image and text" and measuring the ASR as the fraction of responses rating the pair as

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Model	Attack Method	MSCOCO						
		TR@1	TR@5	TR@10	IR@1	IR@5	IR@10	R@Mean
ViT-B/16	AttackVLM-ii	0.4	1.0	1.4	0.24	1.08	2.16	1.05
	AttackVLM-it	0.2	1.4	1.8	0.16	1.16	2.12	1.14
	SASD-WS-Cos	6.0	17.0	24.8	9.08	24.39	34.55	19.30
	SASD-WS-MSE	4.8	18.4	25.6	8.20	25.87	35.15	19.67
	SU-Cos	6.8	20.4	27.8	11.11	25.70	33.34	20.86
	SU-MSE	6.8	20.6	27.0	10.83	25.10	32.62	20.49
	AnyAttack-Cos	8.6	21.2	29.6	10.80	27.59	37.50	22.55
	AnyAttack-Bi	12.2	26.2	33.8	12.63	31.71	40.86	26.23
	AnyAttack-Cos w/ Aux	8.4	24.8	33.0	11.59	32.10	44.98	25.81
	AnyAttack-Bi w/ Aux	**14.8**	**36.8**	**48.0**	**17.59**	**42.02**	**56.05**	**35.88**
ViT-L/14	AttackVLM-ii	0.2	1.0	1.6	0.24	0.60	1.32	0.83
	AttackVLM-it	0.4	0.8	1.4	0.12	0.76	1.48	0.83
	SASD-WS-Cos	3.8	11.6	18.8	7.20	18.43	26.35	14.36
	SASD-WS-MSE	5.4	14.6	20.6	6.00	18.23	26.47	15.22
	SU-Cos	3.0	10.4	13.2	6.19	14.99	20.07	11.31
	SU-MSE	3.4	11.2	17.4	6.63	15.27	19.79	12.28
	AnyAttack-Cos	3.8	14.0	22.8	7.36	20.71	27.55	16.04
	AnyAttack-Bi	4.8	16.0	23.6	8.20	22.31	29.11	17.34
	AnyAttack-Cos w/ Aux	9.4	24.6	37.0	11.51	32.62	48.18	27.22
	AnyAttack-Bi w/ Aux	**12.0**	**34.0**	**47.4**	**15.67**	**39.34**	**53.54**	**33.66**
ViT-L/14 × 336	AttackVLM-ii	0.2	0.6	1.6	0.16	1.12	2.04	0.95
	AttackVLM-it	0.2	0.6	1.8	0.32	0.96	1.76	0.94
	SASD-WS-Cos	2.8	10.8	16.4	6.52	18.31	26.19	13.50
	SASD-WS-MSE	4.4	13.6	19.2	6.72	18.23	25.71	14.64
	SU-Cos	2.4	8.0	11.2	4.88	13.79	18.39	9.78
	SU-MSE	3.6	8.2	13.2	6.40	14.51	19.19	10.85
	AnyAttack-Cos	4.6	11.0	16.6	5.96	17.67	24.23	13.34
	AnyAttack-Bi	3.6	14.4	19.0	7.64	19.79	26.83	15.21
	AnyAttack-Cos w/ Aux	9.0	23.2	37.2	11.68	34.03	47.62	27.12
	AnyAttack-Bi w/ Aux	**12.0**	**33.2**	**46.8**	**14.79**	**39.22**	**53.06**	**33.18**

Table 2. The retrieval performances on the MSCOCO dataset under different attacks. TR@1, TR@5, and TR@10 measures text retrieval performance, while IR@1, IR@5, and IR@10 measures image retrieval performance. R@Mean is the average of all retrieval metrics. Our proposed methods are *italicized*, the best results are highlighted in **bold**, and the second-best results are underlined.

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Attack Method	**Accuracy**
AttackVLM-ii	6.5
AttackVLM-it	6.3
SASD-WS-Cos	24.3
SASD-WS-MSE	24.8
SU-Cos	13.7
SU-MSE	13.6
AnyAttack-Cos	17.5
AnyAttack-Cos w/ Aux	**44.8**

Table 3. Attack performance comparison on the SNLI-VE dataset for multimodal classification.

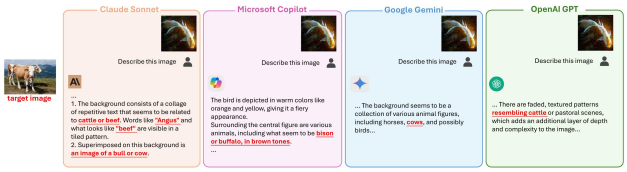

Figure 3. Example responses from commercial VLMs (Gemini 1.5 Flash and GPT-4o mini) to targeted adversarial images generated by AnyAttack. The model incorrectly associates the adversarial image with the attacker-specified target text.

"highly/partially relevant."

As shown in Table 5, *AnyAttack-Cos w/ Aux* achieves ASRs of 31% on Google Gemini 1.5 Flash and 38% on OpenAI GPT-4o mini, dramatically outperforming all baselines (Gemini: $\leq$5%; GPT-4o: $\leq$28%). Figure 3 shows qualitative examples of adversarial images successfully fooling Gemini and GPT-4o into associating the carrier image with the target text.

Model	Attack Method	SPICE	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
InstructBLIP	AttackVLM-ii	1.4	38.9	5.4	8.7	28.6	3.4
	AttackVLM-it	1.3	39.1	5.4	8.7	28.8	3.3
	SASD-WS-Cos	3.4	43.9	7.2	10.5	32.2	10.9
	SASD-WS-MSE	3.2	44.6	7.0	10.8	32.4	11.8
	SU-Cos	1.9	40.7	6.0	9.3	29.9	5.3
	SU-MSE	1.9	40.9	6.5	9.5	29.9	6.0
	AnyAttack-Cos	2.3	41.5	5.9	9.5	30.2	7.0
	AnyAttack-Cos w/ Aux	**4.7**	**46.5**	**7.5**	**12.2**	**33.6**	**20.3**
BLIP2	AttackVLM-ii	1.2	39.6	5.3	8.7	29.0	3.6
	AttackVLM-it	1.2	39.6	5.4	8.7	29.3	3.5
	SASD-WS-Cos	2.6	43.0	6.3	10.2	32.0	9.3
	SASD-WS-MSE	2.8	42.8	6.5	10.2	31.7	9.5
	SU-Cos	1.6	40.9	5.6	9.2	30.1	4.7
	SU-MSE	1.6	40.8	5.9	9.2	30.1	5.0
	AnyAttack-Cos	1.8	41.3	5.2	9.6	30.9	5.6
	AnyAttack-Cos w/ Aux	**3.3**	**44.2**	**6.0**	**11.0**	**32.4**	**13.3**
BLIP	AttackVLM-ii	1.3	39.8	5.0	8.8	29.9	3.4
	AttackVLM-it	1.2	39.7	4.8	8.7	29.7	3.2
	SASD-WS-Cos	3.3	43.8	6.9	10.7	32.3	11.9
	SASD-WS-MSE	**3.4**	43.8	6.9	10.8	32.3	12.4
	SU-Cos	2.6	43.0	6.5	10.1	31.8	8.7
	SU-MSE	2.6	42.4	6.4	9.9	31.6	8.4
	AnyAttack-Cos	2.2	41.6	6.0	9.5	31.1	6.1
	AnyAttack-Cos w/ Aux	**3.4**	**44.4**	**7.1**	**11.1**	**32.8**	**13.0**
Mini-GPT4	AttackVLM-ii	1.6	29.5	2.3	9.3	24.3	1.6
	AttackVLM-it	1.5	29.2	2.3	9.4	24.5	1.5
	SASD-WS-Cos	2.8	30.5	2.4	10.9	25.2	2.6
	SASD-WS-MSE	3.1	30.5	2.9	10.9	25.7	2.8
	SU-Cos	2.0	29.5	2.9	9.9	24.8	1.9
	SU-MSE	2.2	30.3	2.9	9.9	25.1	2.2
	AnyAttack-Cos	2.5	30.5	2.4	10.3	25.2	1.9
	AnyAttack-Cos w/ Aux	**4.6**	**32.5**	**4.0**	**12.4**	**27.3**	**5.3**

Table 4. Attack performance comparison on the MSCOCO dataset for image captioning task.

Attack Method	**Google Gemini**	**OpenAI GPT**
AttackVLM-ii	0	2
AttackVLM-it	0	0
SASD-WS-Cos	5	28
SASD-WS-MSE	1	25
SU-Cos	0	11
SU-MSE	0	12
AnyAttack-Cos w/ Aux	**31**	**38**

Table 5. Quantitative performance comparison on commercial VLMs. The reported values represent the ASR.

4.6. Efficiency Analysis

Figure 4 compares the memory usage and runtime of AnyAttack against baseline methods. Iterative optimization-based attacks (AttackVLM, SU, SASD) require hundreds of gradient steps per sample, resulting in significantly higher memory usage and runtime. AnyAttack generates adversarial perturbations with a single forward pass, achieving order-of-magnitude speedups in inference time and dramatically lower GPU memory requirements. This efficiency makes AnyAttack the only method feasible for attacking large-scale image databases in practice.

4.7. Ablation Studies

Figure 5 presents ablation results on the MSCOCO image-text retrieval task, evaluating the impact of each component. Key findings include:

Pre-training is critical. A model fine-tuned from scratch (without pre-training on LAION-400M) performs substantially worse than one initialized from the pre-trained weights. This confirms that large-scale pre-training instills generalizable adversarial features that are not recoverable from task-specific fine-tuning alone.

Auxiliary encoders improve transferability. Using auxiliary encoders (ViT-B/16 and EVA-02) during fine-tuning consistently improves R@Mean, with gains of up to $\sim$16 points for the larger CLIP models.

Loss function matters. For retrieval, the bidirectional

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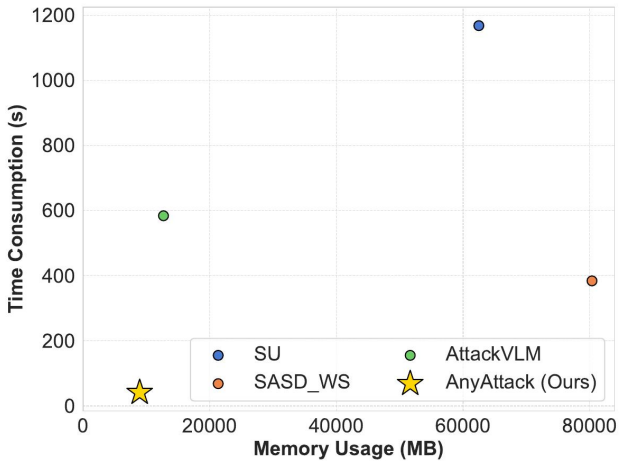

Figure 4. Comparison of memory usage and time consumption across different attack methods. AnyAttack requires only a single forward pass, resulting in substantially lower resource usage compared to iterative baselines.

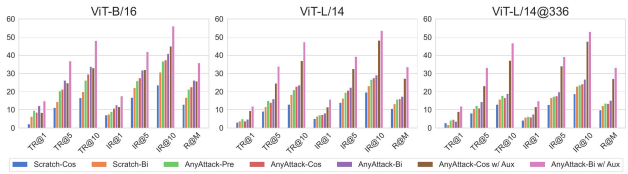

Figure 5. Ablation study: performance comparison between different AnyAttack configurations (with/without pre-training, with/without auxiliary encoders, different loss functions) on the MSCOCO image-text retrieval task.

contrastive loss outperforms the cosine loss, while for classification the cosine loss is more appropriate.

Temperature annealing accelerates convergence. Ablating the exponential temperature schedule reveals that annealing τ from 1.0 to 0.07 over pre-training substantially outperforms using a fixed temperature, as gradual sharpening first captures coarse cross-modal alignment before refining to fine-grained discrimination.

K-augmentation diversifies training pairs. We ablate the number of augmentations $K \in \{1, 3, 5, 10\}$ during pre-training and find that R@Mean improves consistently as K increases from 1 to 5, with diminishing returns beyond $K = 5$. Larger K creates more diverse carrier–target pairings within each mini-batch, encouraging the generator to learn perturbations that are robust to variations in carrier image content and target distribution, without requiring any additional labeled data.

5. Discussion

Why does AnyAttack transfer? The strong cross-model transferability of AnyAttack can be attributed to three factors. First, pre-training on the large-scale LAION-400M dataset forces the generator to learn perturbations that exploit generic visual features shared across VLMs, rather than model-specific artifacts. Second, the InfoNCE contrastive objective encourages the adversarial embedding to be discriminatively close to the target across a large negative pool, producing perturbations that are robust to embedding space variations across models. Third, ensemble supervision with diverse auxiliary encoders during fine-tuning explicitly encourages architecture-independent perturbations.

Security Implications. The success of AnyAttack on commercial VLMs—including closed-source systems accessed only via API—highlights a fundamental vulnerability in current VLM deployments. Unlike jailbreaking attacks [27, 33, 40] that require white-box or iterative black-box access, AnyAttack requires only a single image embedding from the target and can generate adversarial perturbations without querying the target system at all. This underscores the need for adversarial robustness to be considered as a first-class property in VLM training and deployment.

Carrier Image Independence. A practically important property of AnyAttack is its robustness to the choice of carrier image x_r at inference time. Empirically, attack success rates remain stable across diverse carrier sources—including random natural images, synthetic textures, and nearly uniform backgrounds—indicating that the perturbation generator encodes adversarial information directly into δ rather than exploiting specific carrier image statistics. This independence is desirable for real-world deployment, as the attacker requires no curated carrier set, and it further distinguishes AnyAttack from patch-based methods whose effectiveness can depend heavily on the base image.

Limitations. AnyAttack's attack success rates, while substantially higher than baselines, remain below 50% in most settings, suggesting that current VLMs have some inherent resistance to embedding-space attacks. Our approach currently targets image-side perturbations only; incorporating text perturbations in a unified framework is an interesting future direction. Additionally, the perturbation bound $\varepsilon = 16/255$ may be perceptible in some settings. Investigating adversarial training and certified defenses specifically tailored to embedding-space attacks is a natural next step. We also note that attack success rates on commercial VLMs, while far exceeding all baselines, still leave room for improvement, likely due to unknown and potentially more robust internal representations in closed-source systems.

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6. Conclusion

We presented AnyAttack, a self-supervised framework for large-scale targeted adversarial attacks on vision-language models. By reformulating the attack as a foundation model pre-training problem, AnyAttack eliminates the need for explicit target supervision and enables efficient single-pass adversarial example generation. Extensive experiments across multiple datasets, tasks, and VLMs—including commercial systems—demonstrate that AnyAttack substantially outperforms all existing baselines while being significantly more efficient. We hope that AnyAttack will serve as a useful tool for understanding and improving the adversarial robustness of VLMs.

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